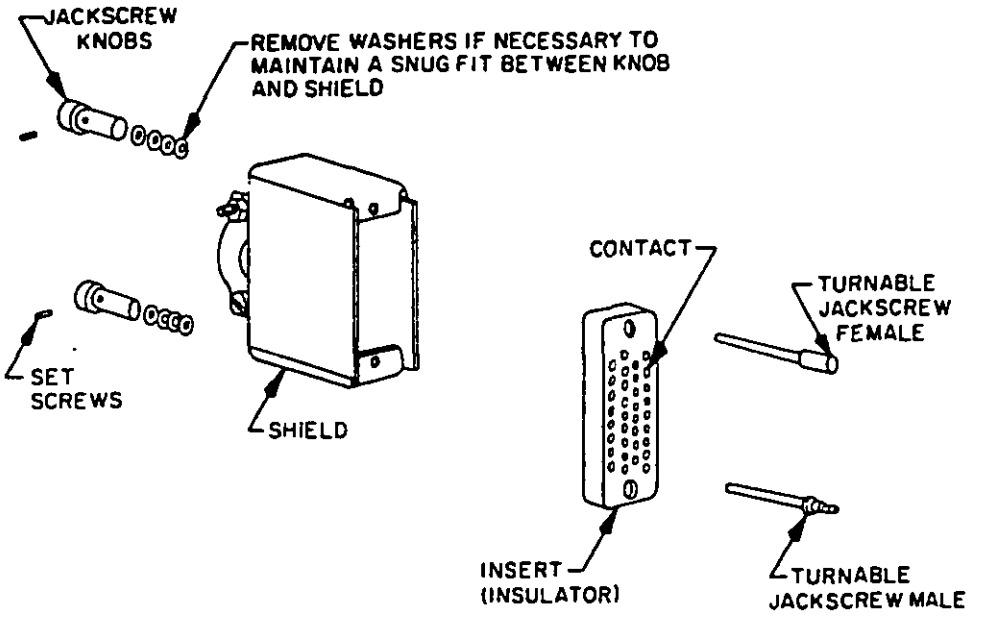
MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR,
RACK AND PANEL, CRIMP TYPE REMOVABLE SOCKET CONTACTS, SIZE 22

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-28748.

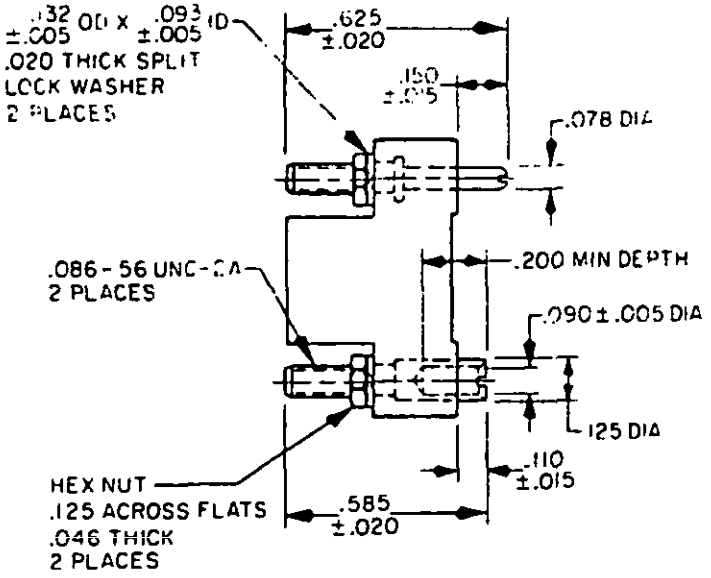


Configuration A. Connector.

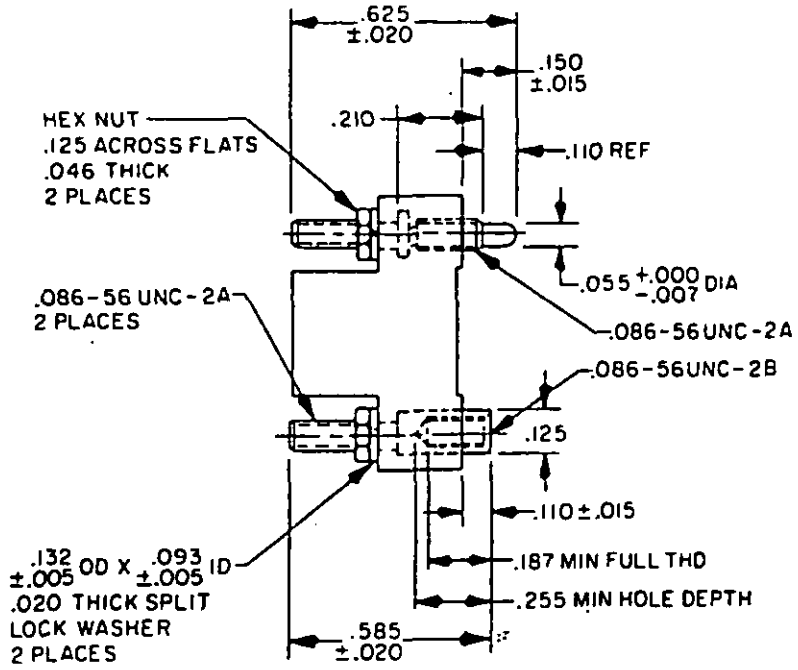
FIGURE 1. Connector with guide pins, jackscrews, and shields.

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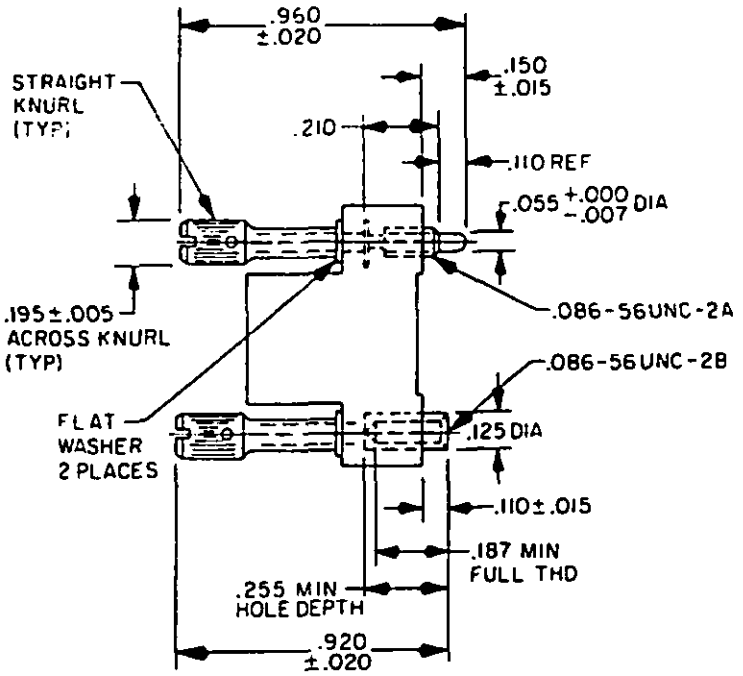
Configuration B. Connector with guide pins.



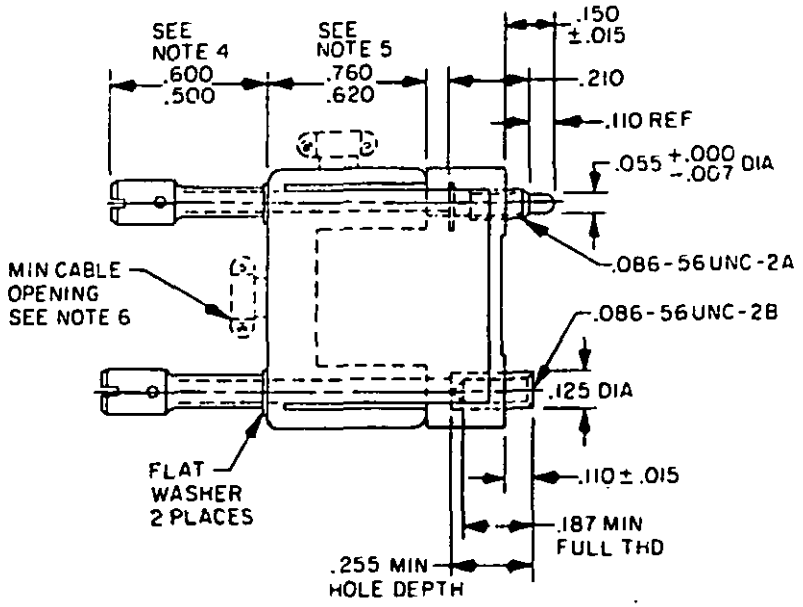
Configuration C. Connector with fixed jackscrews.

FIGURE 1. Connector with guide pins, jackscrews, and shields - Continued.

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Configuration D. Connector with short turnable jackscrews.



Configuration E. Connector with long turnable jackscrews and shield.

FIGURE 1. Connector with guide pins, jackscrews, and shields - Continued.

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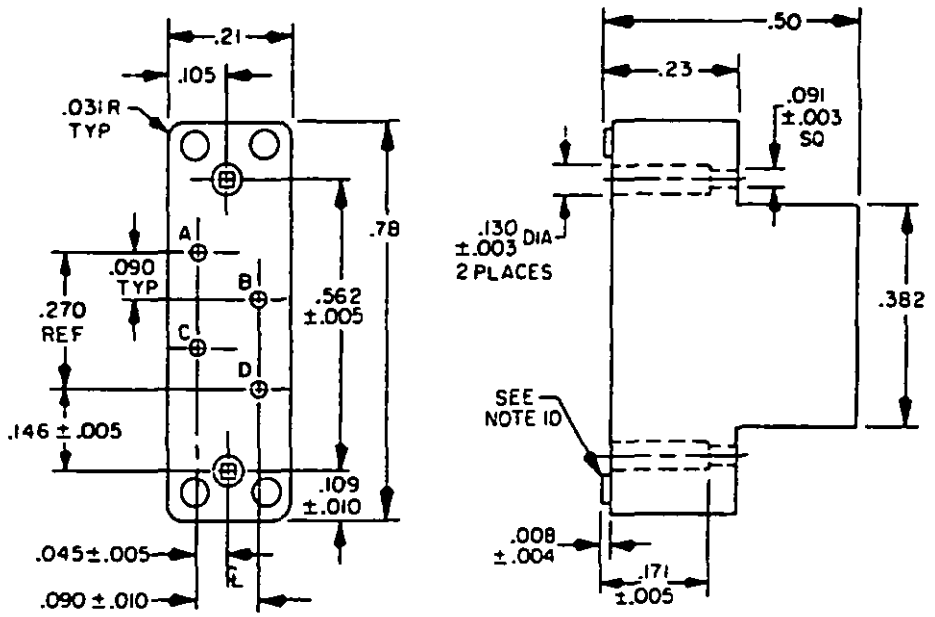
INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.005	.13	.086	2.18	.187	4.75	.600	15.24
.007	.18	.090	2.29	.195	4.95	.620	15.75
.015	.38	.093	2.36	.200	5.08	.625	15.88
.020	.51	.110	2.79	.210	5.33	.700	17.78
.046	1.17	.125	3.18	.255	6.48	.920	23.37
.055	1.40	.132	3.35	.500	12.70	.960	24.38
.078	1.98	.150	3.81	.585	14.86		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.003 (.08 mm).
4. .730 (18.54 mm) maximum for 14 contact connector with side opening shield.
   .630 (16.00 mm) maximum for 50 contact connector.
5. 1.20 (30.48 mm) maximum for 44 contact arrangements.
   1.52 (38.61 mm) maximum for 50 contact arrangements.
6.

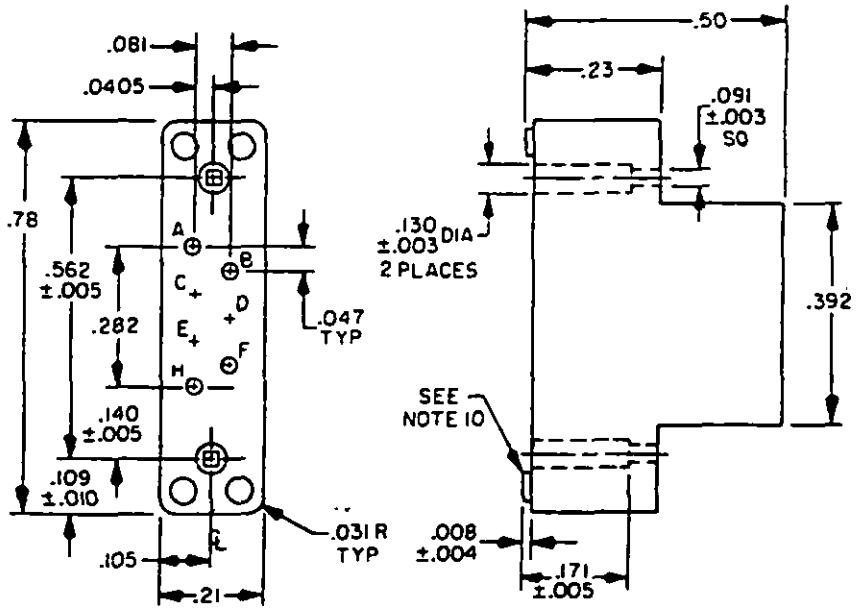
Number of contacts	Minimum cable opening
4	.07 sq. inch
7	.07 sq. inch
14	.19 sq. inch
20	.22 sq. inch
26	.26 sq. inch
34	.34 sq. inch
44	.34 sq. inch
50	.34 sq. inch

FIGURE 1. Connector with guide pins, jackscrews, and shields - Continued.

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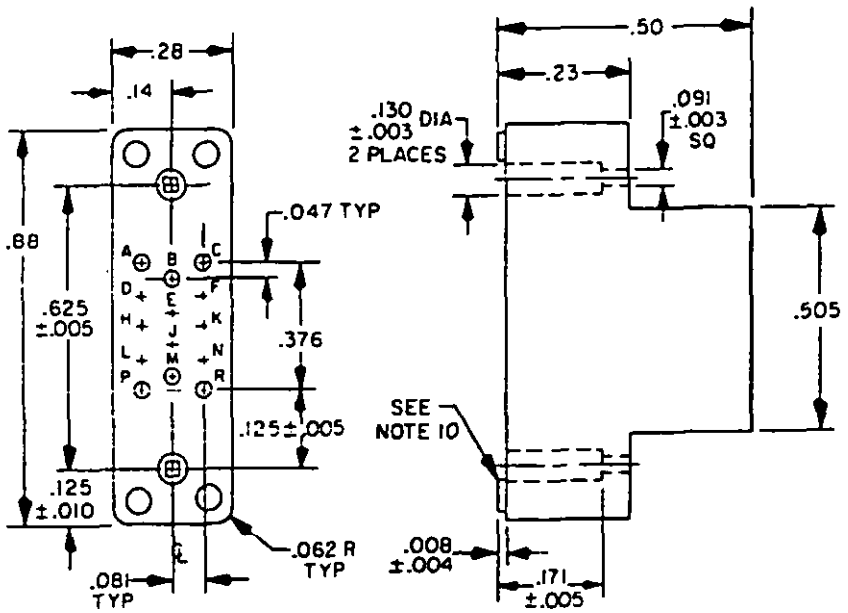
Configuration A. Insert with four contacts.



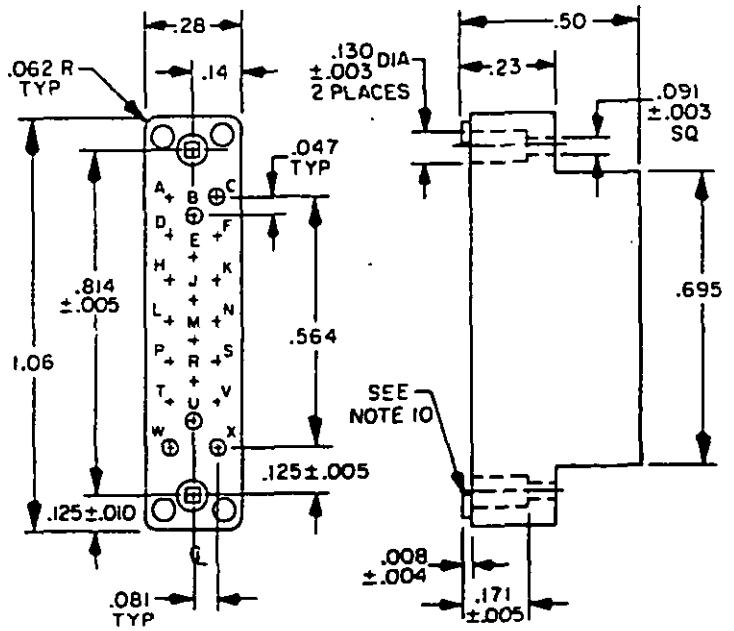
Configuration B. Insert with seven contacts.

FIGURE 2. Insert arrangements.

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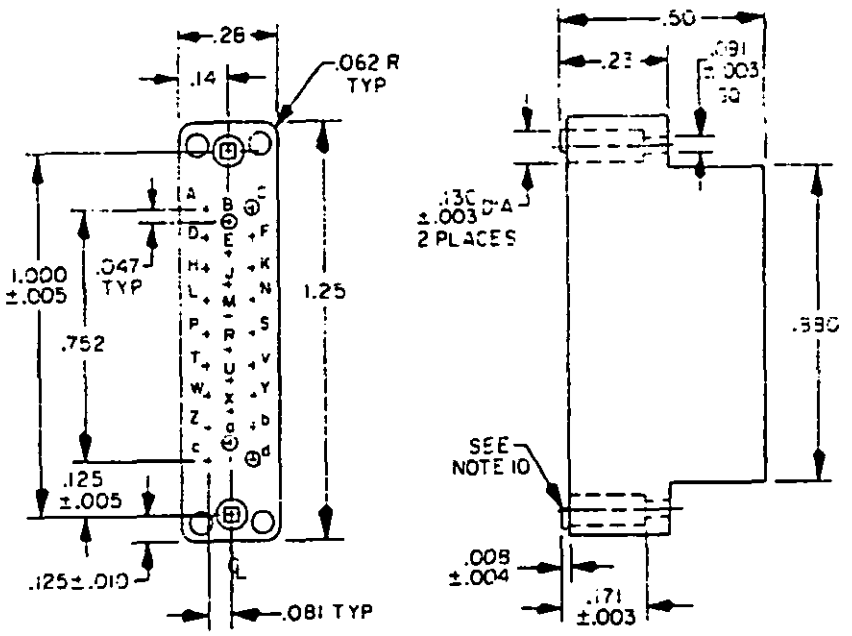
Configuration C. Insert with 14 contacts.



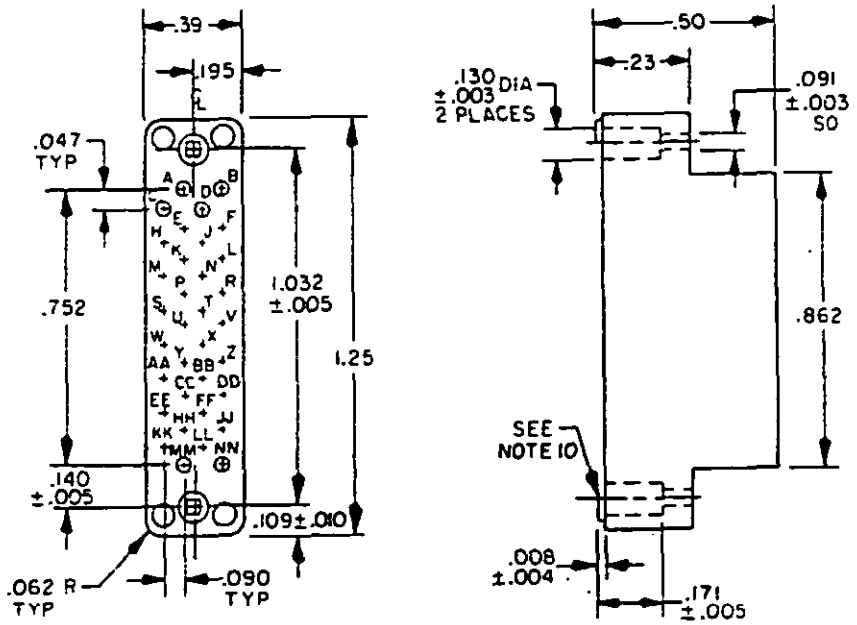
Configuration D. Insert with 20 contacts.

FIGURE 2. Insert arrangements - Continued.

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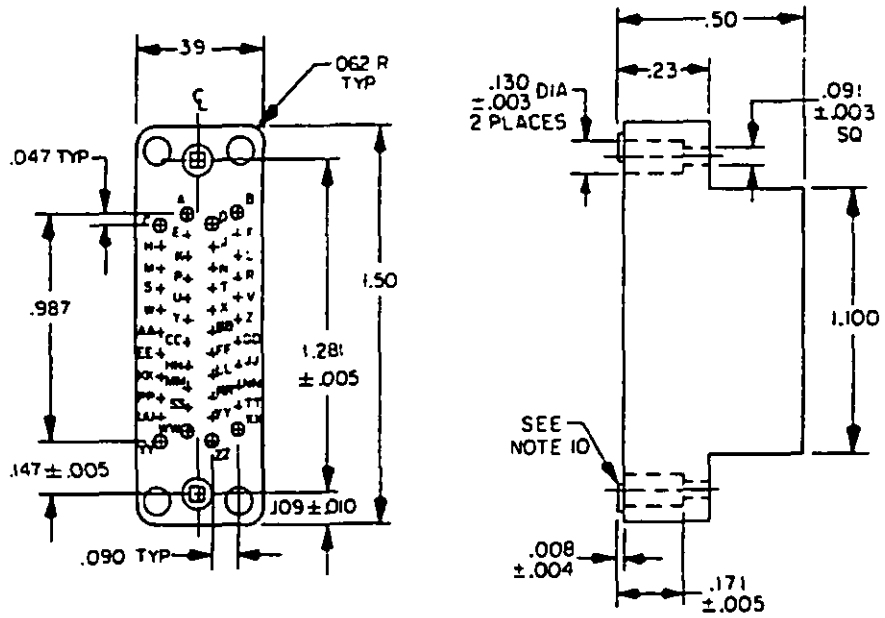
Configuration E. Insert with 26 contacts.



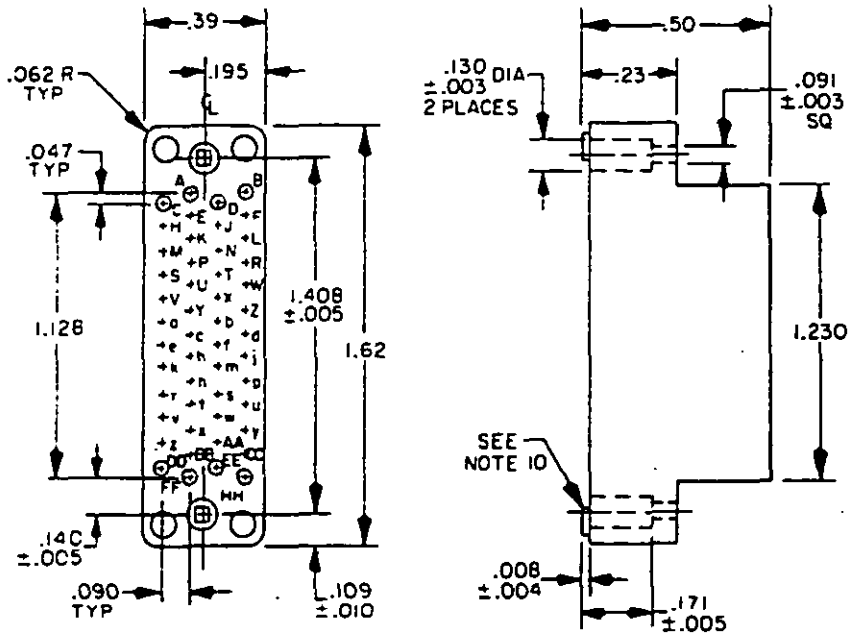
Configuration F. Insert with 34 contacts.

FIGURE 2. Insert arrangements - Continued.

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Configuration G. Insert with 44 contacts.



Configuration H. Insert with 50 contacts.

FIGURE 2. Insert arrangements - Continued.

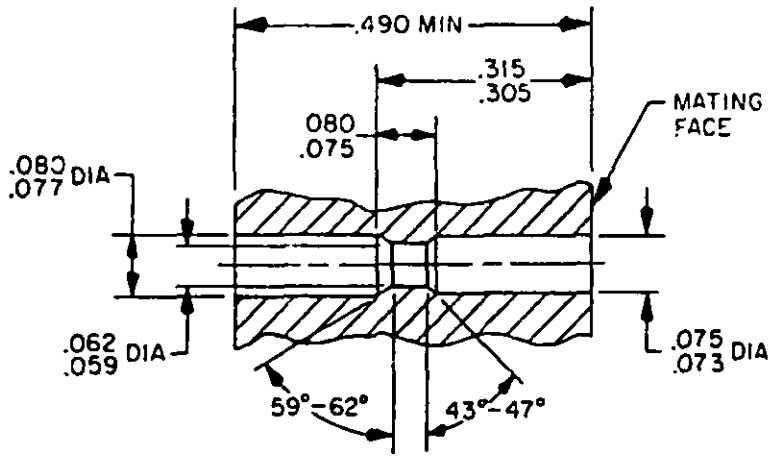
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:18Z
Check the source to verify that this is the current version before use.

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.090	2.29	.23	5.8	.625	15.88	1.100	27.94
.004	.10	.091	2.31	.270	6.86	.695	17.65	1.128	28.65
.005	.13	.105	2.67	.28	7.1	.752	19.10	1.230	31.24
.008	.20	.109	2.77	.282	7.16	.78	19.8	1.25	31.8
.010	.25	.125	3.18	.376	9.55	.814	20.68	1.281	32.54
.031	.79	.130	3.30	.382	9.70	.862	21.89	1.408	35.76
.0405	1.029	.140	3.56	.39	9.9	.88	22.4	1.50	38.1
.045	1.14	.146	3.71	.50	12.7	.987	25.07	1.62	41.1
.047	1.19	.171	4.34	.505	12.83	1.000	25.40		
.062	1.57	.195	4.95	.562	14.27	1.032	26.21		
.081	2.06	.21	5.3	.564	14.33	1.06	26.9		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.02 (.51 mm) for two place decimals and ±.015 (.38 mm) for three place decimals.
4. Tolerance between any two adjacent contact centers shall be ±.004 (.10 mm). Tolerance between any two contact centers, other than adjacent contacts, shall be ±.006 (.15 mm).
5. Contact letter identification may be located on either side of the contact hole. All letters shall appear on the front and rear face of each insert.
6. Inserts shall be used with guide pin assemblies or jackscrew assemblies.
7. Insert material: Shall be in accordance with MIL-C-28748.
8. Installing and removal tools: MIL-T-81969/18 and /20.
9. Crimp tool: M22520/2-01. Positioner: Buchanan 615695, Daniels K280, or equivalent.
10. Shape of .008 pads optional.
11. Mating face of socket insert shown; pin insert is reverse image.
12. Contacts: MIL-C-39029/35.

FIGURE 2. Insert arrangements - Continued.

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INCHES	MM
.059	1.50
.062	1.57
.073	1.85
.075	1.90
.077	1.96
.080	2.03
.305	7.75
.315	8.00
.490	12.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 3. Insert hole configuration.

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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2, and 3.

Mating connectors: MIL-C-28748/7 and MIL-C-28748/13.

Resistance to test probe damage .145 ±.015 depth of insertion.

Part number: As shown in the following example:

EXAMPLE OF PART NUMBER:

(See part number code)

M28748/ 14	A	0	0	F	1A
Basic number of specification sheet	Insert ARR. (Fig. 2)	Shield and shield clamp location or retaining plate (Fig. 1)	Shell type	Jackscrews or guidepins (Fig. 1)	Contacts

PART NUMBER CODE:

Insert arrangement figure 2 configuration

A -	4
B -	7
C -	14
D -	20
E -	26
F -	34
G -	44
H -	50

Number of contacts

Shield and shield clamp location or retaining plate

Shield
1-Top opening per configuration E of figure 1
2-Side opening per configuration E of figure 1
0-None included

Shell type

0 - None included

Jackscrews or guidepins

L - Long jackscrews: Conf E, figure 1
F - Fixed jackscrews: Conf C, figure 1
S - Short jackscrews: Conf B, figure 1
G - Guidepins: Conf B, figure 1

Jack and guide location: The jackscrew (turning or fixed) or guidepin shall be located by the "A" socket contact.

Contacts: Per MIL-C-39029/35.
1A - 100 percent size 22

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, AT
Navy - AS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AV, MC
Navy - MC, OS
Air Force -

Preparing activity:
Navy - EC

Agent:
DLA - ES

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A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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